Lab 09 -Functions

Objective

Solving exercises from the textbook in chapter 2.3

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. will understand functions and different types of functions
2. Finding the domain and range of function, composite of function and graph of function.
3. Will be able to solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab and use blackboard slides in order to solve the exercises.

Lab Assessment

1- Divide students to groups and let them to solve the given example.
2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

In this lab, the following exercises are going to be solved and explained to them:

1. Why is f not a function from R to R if
   a) f (x) = 1/x?
   b) f (x) = $\sqrt{\ }$x?

2. Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.

a) the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string
b) the function that assigns to each bit string twice the number of zeros in that string
c) the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)

3. find the domain of the function f(x)=x^2+3x+5/x^2-5x+4.

4. find the domain and range of the function f(x)=x-2/3-x

5. Find these values.
   a)$\lceil 3/4 \rceil$ b) $\lceil -3/4 \rceil$ c)$\lfloor -1 \rfloor$ d) $\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor$ e) $\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor$

6. Consider these functions from the set of teachers in a school.
   Under what conditions is the function one to one if it assigns to a teacher his or her
   a. Office
   b. Salary
   c. Social security number

7. Determine whether the function $f: \mathbf{Z} \times \mathbf{Z} \to \mathbf{Z}$ is onto if

   a) $f(m, n) = m + n$. **b)** $f(m, n) = m^2 + n^2$. **c)** $f(m, n) = m$.

   d) $f(m, n) = |n|$. **e)** $f(m, n) = m - n$.

8. Determine whether each of these functions is a bijection from **R** to **R**.

a) $f(x) = 2x + 1$ **b)** $f(x) = x^2 + 1$ **c)** $f(x) = x^3$ **d)** $f(x) = (x^2 + 1)/(x^2 + 2)$

9. Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if

a) $S = \{-2, -1, 0, 1, 2, 3\}$. **b)** $S = \{0, 1, 2, 3, 4, 5\}$. **c)** $S = \{1, 5, 7, 11\}$. **d)** $S = \{2, 6, 10, 14\}$.

10. How many bytes are required to encode n bits of data where n equals
    a) 7? **b)** 17? **c)** 1001? **d)** 28,800?

11. Draw the graph of the function $f(x) = \lfloor x \rfloor + \lfloor x/2 \rfloor$ from **R** to **R**.
12. Given $f(x) = 2x + 3$ and $g(x) = -x^2 + 5$, find $(f \circ g)(x)$, $(g \circ f)(x)$.
13. Let $f(x) = -x + 4$, find $f^{-1}(x)$. $f: R \to R$ b) $f(x) = x^3 + 1$, $f: R \to R$